Global Facilities - Design & Construction Standard - CSA - Cast-In-Situ Micro Piles Specifications

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ECN History **Submit a Change Request**

1.0 Purpose

1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Cast-In-Situ Micro Piles as part of any green and/or brown field construction projects contract requirement.

1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.

1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.

1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:

1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing

1.4.2 Environmental, Health & Safety Standards

1.4.3 Operational Requirements /SOP (Standard Operation Procedures)

1.4.4 Security Protocol

1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.

1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

2.1 This document for CIS micro piles specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
AE	External Architectural & Engineering Consultants
CIS	Cast-In-Situ
DPT	Micron's Designated Project Team
EHS	Micron's Environmental Health & Safety Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 General

4.1 Design

All materials and work shall be in accordance with drawings, this specification and all current Codes of Practice, Euro codes, SS, or equivalent local/international code requirements.

Notwithstanding the above, tests shall be carried out as and when directed by the Engineer in accordance with the relevant standards.

4.2 CODE OF PRACTICES

BS EN 1997-1 Geotechnical design - Part 1: General rules

BS EN 1997-2 Geotechnical design - Part 2: Ground investigation and testing

Or equivalent local/international code requirements in the project location.

4.3 The piles shall be capable of carrying the specified working load with a geotechnical safety factor of 2.5.

4.4 Due to soil disturbances, the top 1.5 metres of micro-piles shall be excluded in the calculation of effective length of piles.

4.5 The Contractor carrying out the works shall be a qualified foundation Specialist who has the proven records in the design, supply and installation of micro-piles locally.

4.6 The Contractor shall submit design calculations and details of micro-piles for the Engineer's concurrence prior to commencement of works.

4.7 Such calculations and details shall be prepared and certified by the Contractor's Professional Engineer, and the Contractor shall make such amendments and changes to the design/details deemed necessary by the Project Engineer without additional charge.

4.8 The Contractor shall also provide Professional Engineer certified final calculations and details of the proposed micro-pile system including capping details etc. for submission to the Building Authority for approval and shall alter/amend such calculations/drawings to comply with the requirements of the Authority without additional cost.

5.0 System of Piling

The piling system and equipment proposed by the contractor.

5.1 Micro-piles may be formed either by boring, placing an axial steel component and grouting (Bored Micropiles) or driving patented steel tubes (Driven Micropiles).

5.2 The system and equipment shall be suitable for all type of soil and rock formations. Use appropriate drilling bits, including down-the-hole hammer for rock formation.

5.3 The system and equipment's used shall be capable of working within the constraints of space and headroom at the site. In confined spaces, all equipment shall be operated with external power pack.

5.4 The system and equipment's used shall not induce vibration, which affect the stability of adjacent properties and operation/production of adjacent factories.

6.0 Bored Grouted Micro-Piles

6.1 Bored Grouted Micropiles shall be capable of:

a) Primary Grouting – placed by tremie under low grouting pressure in one continuous operation.

b) Secondary Grouting – injected under high pressure through expandable grout valves embedded in primary grout shaft.

6.2 Materials

6.2.1 Steel Components

- Axial steel component shall be high tensile steel reinforcement bars or preformed standard structural steel sections (such as tubular API pipes or H-sections).
- Steel sectional area shall not exceed 8% in general and 10% at the splice.
- Axial steel component shall include top steel bearing plate, and axial steel component shall extend to the full depth of pile.
- Steel reinforcement shall consist of a minimum of three (3) high tensile reinforcement bars. Bar spacing shall be not less than the bar diameter plus 6mm to allow the grout to flow freely.
- Spiral links shall be provided throughout the length of steel component.
- Jointing of steel components shall be by mechanical means, such as threaded coupler or splice, or approved similar. Test results by approved authority shall be furnished to show that the strength of splices exceeds the specified tensile strength of main component.
- Mechanical threaded joints shall be staggered, and the grout cover shall not be less than 40mm.

6.2.2 Grout

Materials - Grout shall be mixed from ordinary Portland cement complying with current British/Singapore standard and clean water supplied from the public mains.

Grout Admixture - Grout admixture ‘Conbex 100’ or approved similar may be added to manufacturer's recommendation to improve workability and reduce bleeding and shrinkage

Primary Grout Mix - The proportions of grout and the minimum strength of work cubes shall comply with the following requirements in **Table 1**:

Table 1: Primary Grout Mix

Water / Cement Ratio	Resistance to crushing after mixing (Cube Strength)	
	7 days (N/mm^2)	28 days (N/mm^2)
< 0.45	24.0	35.0

The quantities of cement and admixture in the mix shall be measured by weight. The grout shall be free from segregation, slumping and bleeding. Grout shall be mixed on site and shall be pumped into its final position as soon as possible and in no case more than half an hour after mixing. Water and admixtures shall be added to the mixer before the cement.

6.2.3 Fabrication

- For jointing of steel components with threaded coupler, only factory thread bars shall be used.
- Temporary casings shall be used when necessary to prevent collapse of bored-holes. Permanent metal casings shall be provided in soft and loose soils in which `necking' may occur during installation and or grouting.
- When steel bars in clusters are used, spacer devices shall be provided at not more than 3 metres intervals to centralise the steel component in the borehole and to provide a positive grip between the bars and the surrounding grout yet permit the grout to adequately penetrate and completely cover the bars.

7.0 Driven Micro Piles

7.1 Driven micro-piles shall consist of grouted high yield steel tubes complete with approved steel capping plates and mechanical joints.

7.2 The minimum diameter of driven micro piles shall be 75mm internally.

7.3 Piles shall be installed closed-ended with solid mild steel shoes and m.s. dowels.

7.4 After installation, the steel tubes shall be fully grouted with grout mix, as specified in Clauses 3.22 herein

7.5 Capping plates shall have a minimum thickness of 12mm and sufficient surface bearing to transmit the pile load to the pilecaps. A 1m long high yield steel bar of minimum diameter 16mm shall be fillet-welded to the underside of the capping plate.

8.0 Method of Statements

The Contractor shall submit a Method-Statement describing in detail the equipment’s used and complete procedure of installing the micro-piles and capping plates.

9.0 Allowable Tolerances of Installed Piles

9.1 The centroid of each pile shall not deviate by more than 50mm from its true position. The verticality of the micro pile shall not deviate by more than 1 (horizontal) in 75 (vertical) from its vertical position.

9.2 For pile group of three or more micro piles, the positional tolerance of the individual micro piles may exceed above limits. However, the positional tolerance of the centroid of the pile group shall be 75mm.

Any pile or pile group with position or verticality exceeding the above-mentioned limits will be rejected and replaced at the Contractors' costs. Alternatively, the Contractor may propose modifications to the pile cap and/or beam or any remedial works subject to the approval of the Engineer.

10.0 Test To Be Carried Out

The Contractor shall carry out the following tests:

10.1 Grout Test

A maximum of two sets of two test cubes shall be taken from the grout mix for testing purpose, unless further tests are required due to failure in complying with the specified strength.

Two cubes shall be tested at seven days and the remaining two cubes at twenty-eight days after casting.

10.2 Axial Steel Component

(a)Three (3) numbers of tensile tests on steel bars.

11.0 Load Test on Micro Piles

11.1 Standard static working load test shall be carried out in accordance with code requirements.

Testing frequency shall be: Two (2) numbers or 1% of working piles installed or one (1) number for every 50 metres length of proposed building, whichever is greater.

Details of proposal shall be submitted to the Engineer for approval.

11.2 Working load of piles shall be determined from the ultimate load using a minimum factor of safety of 2.5.

11.3 If load tests indicate that the micro piles are not able to achieve the design working load of piles, the Contractor shall submit to the Engineer for approval, proposals on compensating piles.

The additional costs of compensating piles and other incidental costs related to such piles shall be borne by the Contractor.

12.0 As-Built Piling Plans

12.1 On completion of pile installation, and within 14 days after completion, submit the as-built piling plans. (1 soft copy in latest version of AutoCad and 2 paper prints) incorporating pile penetrations, and location and type of pile load tests carried out.

12.2 Also, submit as-built piling plans (1 soft copy in latest version of AutoCad and 2 paper prints, certified by a Licensed Survey) incorporating records of pile eccentricities within 7 days after pile-tops have been exposed.

13.0 Document Control

13.1 Approval: GLOBAL_FAC_GFTT_APPR

13.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_ NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

13.3 Notification:

All Micron Team Members at all Sites

13.4 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

13.5 Review

Biennially (two years)

14.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	07/04/2022